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Key ingredients to avoid in ultra processed foods

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If you're like many people, you've heard the message to cut back on ultra processed foods.

However, not everything in the center of the grocery store raises your risk of heart disease, cancer or type 2 diabetes. That even includes packaged foods with long lists of ingredients you can't pronounce.

"While the overconsumption of ultra processed foods has been linked to adverse health outcomes, that doesn't necessarily mean all processed foods are bad for all people in all situations," says Tara Schmidt, M. Ed., RDN, lead dietitian for the Mayo Clinic Diet.

Some ultra processed foods don't negatively affect health. Others only harm health in high amounts. Still others may protect your health.

What are ultra processed foods?

Processed foods have been altered from their original state.

Sometimes, those changes are beneficial. Take prechopped veggies and frozen chopped fruit. “These slightly modified options contain as many nutrients as their unprocessed counterparts, yet they are easier to prepare,” says Schmidt.

Similarly, commercially prepared hummus can help you consume more fiber. That’s key because most people fall short on this health-promoting nutrient.

So, rather than sorting all processed foods into the “don’t eat” category, think of processing as a continuum that goes from minimally processed to ultra processed, says Schmidt.

The four types of processed foods

Developed several years ago by researchers at the University of São Paulo, Brazil, the NOVA continuum sorts foods into four categories based on their level of processing. At one end of the continuum, you’ll find unprocessed and minimally processed foods like apples and oregano. At the other end are ultra processed foods and beverages, which have the potential to be the most concerning health-wise. These ultra processed options tend to contain:

- Five or more ingredients.
- Added sugar, salt, oils, fats, stabilizers and preservatives.
- Additives that imitate the flavor, color or texture of less processed foods.
- Ingredients not found in nature, such as high fructose corn syrup.

They’re also defined by what they lack. These ready-to-eat products are often so processed that vegetables, grains and other unprocessed foods are either barely present or absent.

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Granted, if you're merely trying to decide which processed foods to put in your cart and which ones to leave on the shelf, the NOVA system can seem confusing. NOVA sorts many flour-based products into group 4 because they often contain the group 2 ingredients sugar and salt, along with emulsifiers that help enhance taste, texture and shelf life. However, whole-grain cereals and breads contain more health-promoting nutrients than those made with refined flour, even though both are listed in the same NOVA category.

The NOVA Food Classification System

Category	GROUP 1 Unprocessed & minimally processed foods	GROUP 2 Processed culinary ingredients	GROUP 3 Processed foods	GROUP 4 Ultra processed foods
Definition	Obtained directly from plants or animals or undergone mild changes (such as freezing)	Includes ingredients either extracted from group 1 foods or nature	Foods created by adding ingredients from group 2 to foods in group 1	Foods created chiefly or entirely from substances in groups 2 and 3
Examples	Fresh, frozen, chopped or dried produce Fresh, sliced and dried mushrooms Pasteurized and powdered milk Spices Unsalted nuts Eggs Dried legumes Whole or chopped potatoes and winter squash	Sea salt Cane and beet sugar Honey Maple syrup Butter Lard Vegetable oils Vinegar Corn starch	Homemade bread Canned fruit, vegetables or legumes Salted nuts Smoked or cured meat Canned tuna Cheese	Ice cream Many snack foods such as potato chips Fish sticks Soft drinks Breakfast cereals and bars Alcohol Commercially baked goods such as cookies and doughnuts

How do ultra processed foods affect your health?

For one review, published in the [British Medical Journal](#), researchers pooled the results from 14 studies involving nearly 10 million people. After crunching the data, the researchers determined that the consumption of ultra processed foods was associated with a higher risk of premature death.[3] Other research has linked high consumption of ultra processed foods with the following health problems:

- Heart disease and heart attacks.
- Stroke.
- High blood pressure.
- Depression.
- Overweight and obesity.
- Diabetes.
- Reduced HDL cholesterol.
- Cancer.

Though much of what you'll find within NOVA's Group 4 category tends to earn F ratings from nutrition researchers, not all foods within that category negatively affect health. As mentioned earlier, according to the NOVA classification system, commercially prepared whole-grain bread falls into the ultra processed bucket. However, [research](#) that looked at the health outcomes of 4,461 people found no association between whole-grain bread and increased risk for cancer or heart disease. To the contrary, it and other whole-grain options likely help to reduce risk for heart disease, diabetes and obesity.

Are some processed ingredients more harmful than others?

As mentioned above, some processed foods can help you eat more essential nutrients. Others, however, contain ingredients or chemicals that may worsen health in the following ways.

Ingredients that raise the risk of cancer

The NOVA food classification system sorts hot dogs, sausages, bacon and other types of processed meat into Group 3. However, these foods may be more harmful to health than some of the ultra processed foods in Group 4. Meat processing via salting, curing, fermentation and smoking can lead to the formation of harmful chemicals called N-nitroso-compounds (NOC) and polycyclic aromatic hydrocarbons (PAH).



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According to [the World Health Organization's International Agency for Research on Cancer](#) (WHO IARC), processed meat is a “known human carcinogen.” That puts these foods in the same cancer-causing category as cigarettes.

Other processed foods with potentially carcinogenic ingredients include:

- Alcohol, which research has linked with mouth, throat, esophagus, colon, liver and breast cancers.
- Fried foods, as high-heat cooking, encourages the formation of acrylamide, acrolein and heterocyclic amines (HCAs), all chemicals thought to raise cancer risk.
- Sugar-sweetened, high-calorie, low-nutrient foods and beverages, which may indirectly raise cancer risk by contributing to weight gain.

Chemicals that disrupt the endocrine system

The industrial chemicals bisphenol A, S and F (BPS, BPF, BPA) are used in some plastic food containers, fast-food wrappers, and the lining of some metal cans and bottle tops. These chemicals can seep into food. Though the U.S. Food and Drug Administration (FDA) has declared bisphenol safe at low levels, other organizations argue that the chemical interferes with hormones such as estrogen and insulin, potentially raising the risk for diabetes and obesity.

Ingredients that promote obesity

Many ultra processed foods contain mixtures of fat, sugar, sodium or starch that make them more appealing, easier to consume quickly and more challenging to stop eating.

How to think about processed foods

Consider the following questions when deciding whether to put packaged food in your grocery cart.

Do the benefits outweigh the risks? Processed foods can present a tradeoff. Some make health-promoting food more convenient, such as filets of fish or pre-washed veggies. During processing, however, other foods may lose health-promoting nutrients or gain substances known to disrupt health.

Take canned beans. They're a convenient and affordable way to consume more health-promoting fiber. However, some may contain added sodium and potentially minute amounts of bisphenol from the lining of the can.

In this example, the benefits far outweigh the risks, says Schmidt, especially if you are rinsing and draining the beans, as this will wash away most sodium and bisphenol.

Similarly, whole-grain bread offers another easy way to bump up your fiber intake. When choosing flour-based foods, look at the ingredient list. Look for the word "whole" in front of the word "wheat." Whole wheat, brown rice, quinoa, spelt, rye and other whole grains should dominate the beginning of the list. Refined grains — such as wheat without the word "whole" — along with additives like sugar and salt should appear toward the end of the list. That means only scant amounts were used. While whole-grain bread is processed, the benefit outweighs the risk due to the predominance of healthy ingredients included.

What's the context? Some of NOVA's group 4 foods offer benefits in some situations, but not in others. For example, sports drinks fall into NOVA's ultra processed category. They don't necessarily make sense for everyone. However, if you compete in long endurance events such as a marathon, these beverages may help you to go the distance without upsetting your stomach.

How often do you consume them? Reducing the intake of something you eat in large portions daily will make more of a health impact than eliminating something you consume rarely. For example, if you want to reduce your added sugar intake, you might reduce or eliminate the three cans of cola you drink daily rather than worry about the dollop or two of ketchup you use once a week, says Schmidt.

What else do you eat? Assuming the bulk of your diet comes from NOVA's group 1 foods — produce, lean protein, healthy fats, legumes and whole grains — you have room for modest portions of ultra processed snack chips and sweets.

Can you moderate it? If you can put a food away after consuming a modest portion, it's likely OK to have some ultra processed treats in the home. On the other hand, if eating one store-bought cookie quickly turns into eating a whole sleeve of cookies, it's likely better to avoid having it in your home.